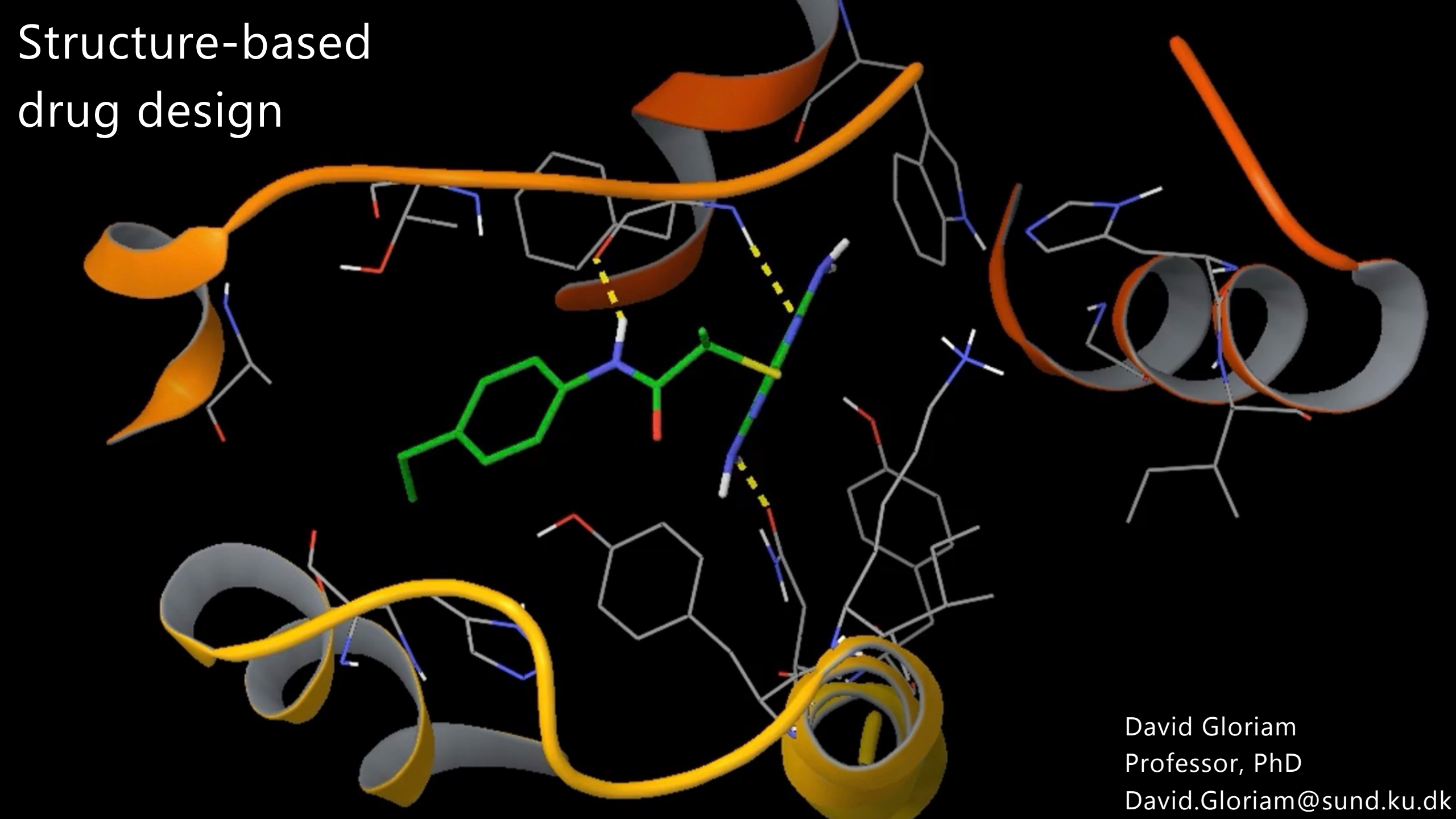
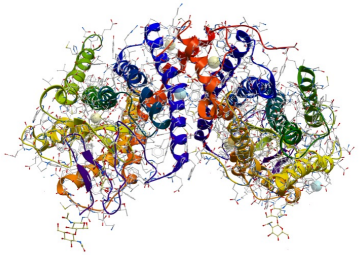
Structure-based drug design



David Gloriam
Professor, PhD
David.Gloriam@sund.ku.dk


David Gloriam
Professor in Computational Receptor Biology
University of Copenhagen

Computational ligand identification and optimization – Integrated in interdisciplinary iterative workflow



Structural biology
Cryo-EM & crystallography

Computational drug design

Chemistry

New drugs

Translational Pharmacology

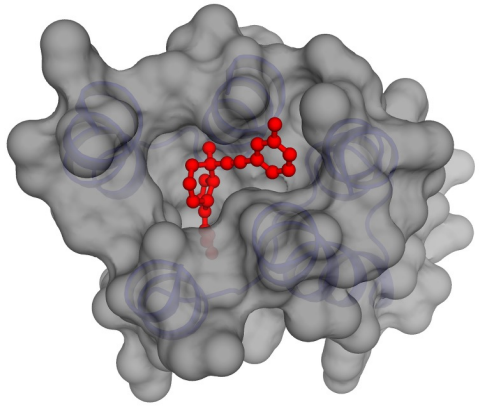
Molecular Pharmacology

Lecture learning goals

- Get to know key methods for structure-based drug design.
- Learn which method to use for which application.

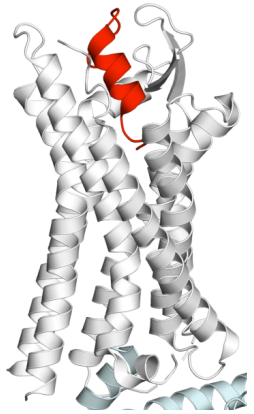
Small molecule ligand

identification and optimization



Protein modeling & peptide ligand

identification and optimization

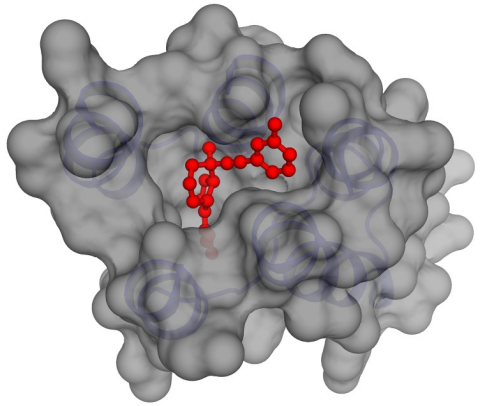


Lecture learning goals

- Get to know key methods for structure-based drug design.
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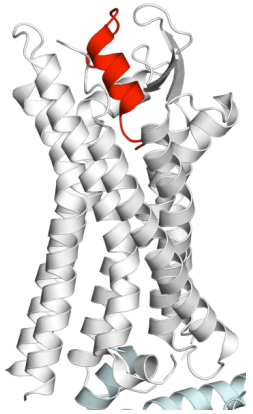
Small molecule ligand

identification and optimization



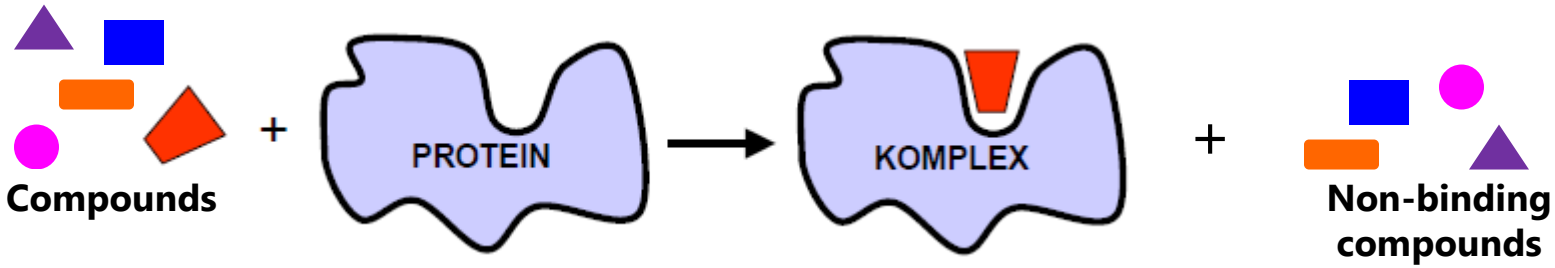
Protein modeling & peptide ligand

identification and optimization

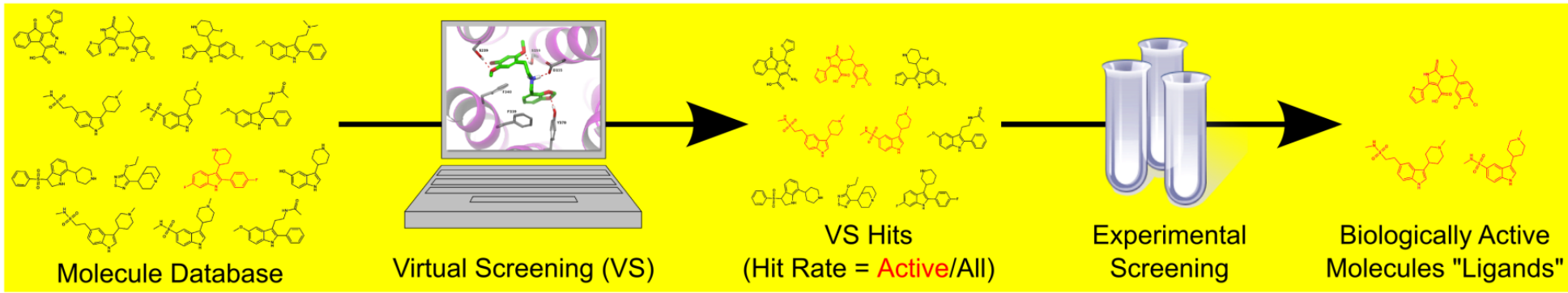


Docking (of molecules into protein target)

Docking simulates binding of ligand into a protein 3D structure

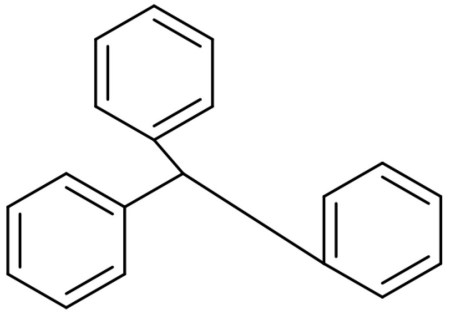


Docking-based virtual screening picks molecules for experiments



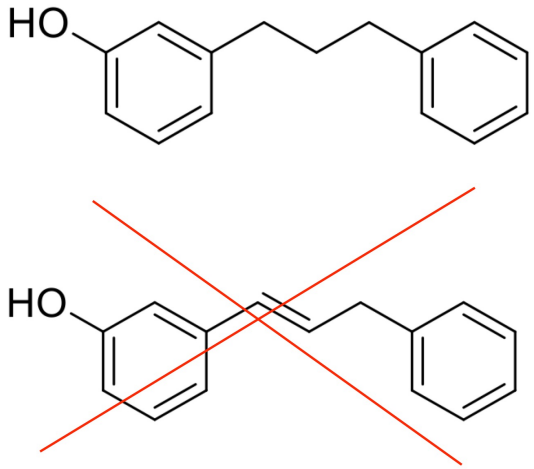
Post-docking filtering of hits

Remove conformationally strained ligands

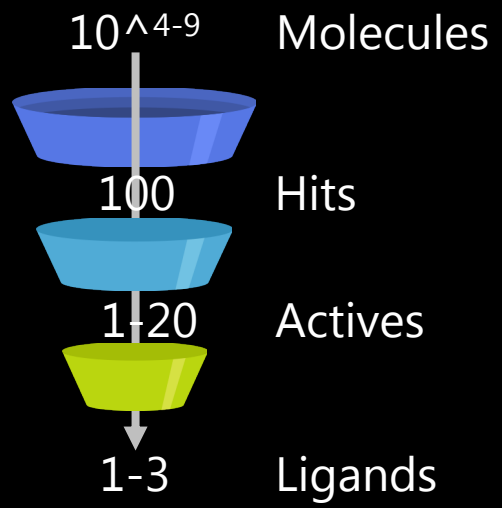


Increase diversity

(by 2D structure clustering)



Virtual screening by docking





Compound databases



The largest single supplier

- Largest single vendor
- 7 Million cmpds in stock
- 150 Billion cmpds made on-demand
- 3.4 Trillion cmpds for analog searches





Meta-databases

- 5 / 20 Million compounds in stock
- Orders compounds from many vendors

Docking has unavoidable limitations

Problem 1



Many targets inaccessible due to requirement of structures

"Dockable" targets

95 Inactive / 213 Active

<<

GPCRs

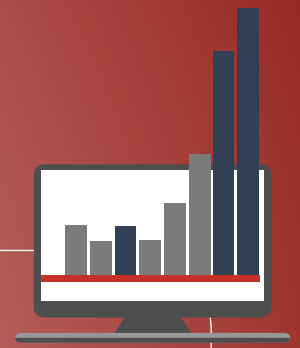
362

Cryo-EM cannot be used on inactive GPCRs (too small)

Many structures cannot be used (wrong state / no ligand)

AlphaFold etc. cannot identify new small molecules

Problem 2



Too slow to explore the huge chemical space

Largest docking

1bn •

<<

Largest database

77bn

Compound databases grow faster than computing power

Docking uses slow physics-based algorithms

AlphaFold etc. are too slow for virtual screening

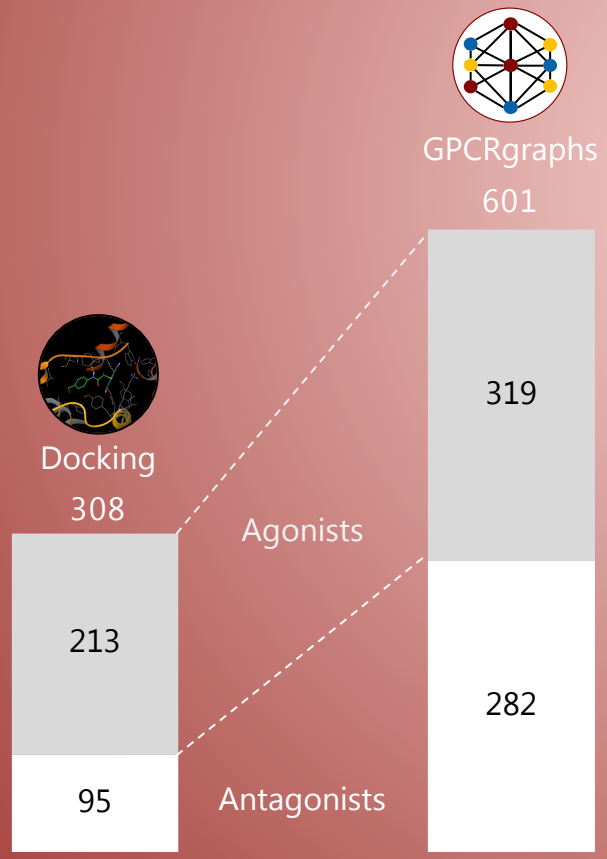
GPCRgraphs solves both problems!



Jascha Henseler

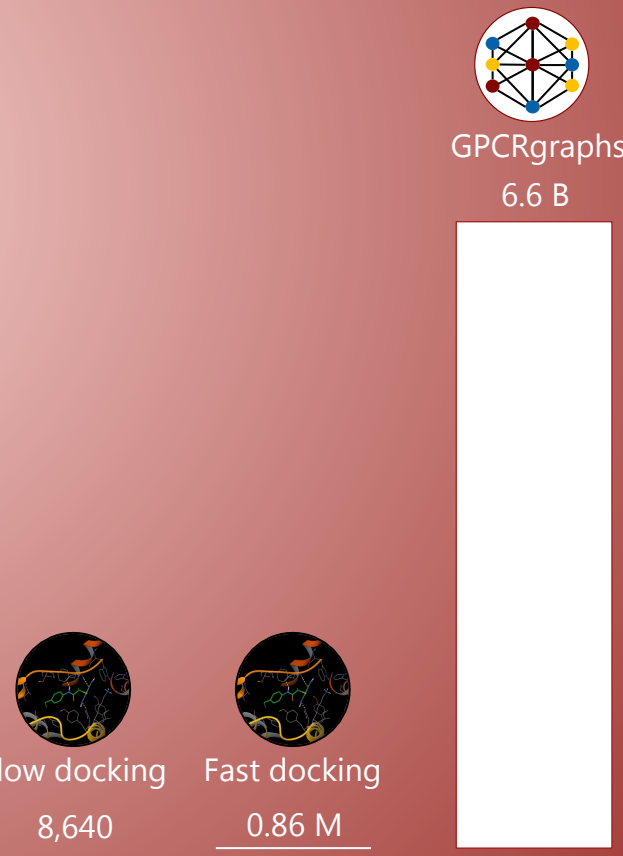
Double as many targets

No structure requirement



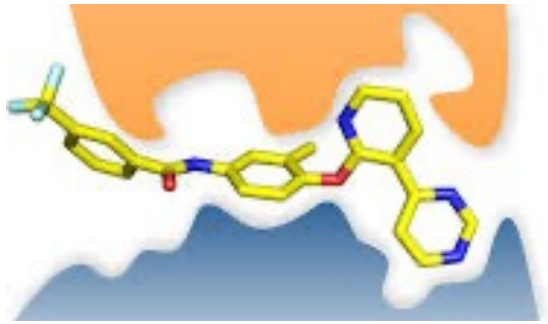
>100,000 times faster

Screened molecules / CPU core and day

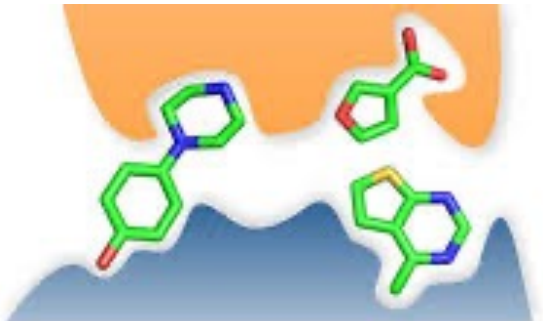


Fragment-based ligand discovery
- Another method for small molecule drug design

Hit from
virtual screening



Hits from
fragment-based discovery

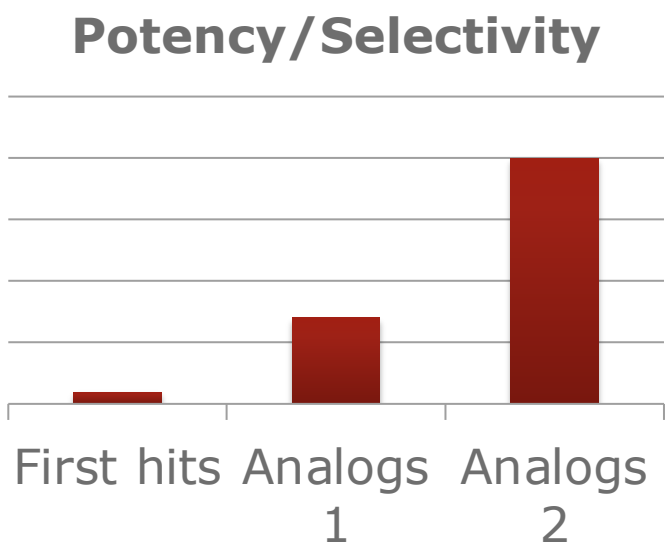


Anders Bach, ILF, UCPH
Runs a fragment facility



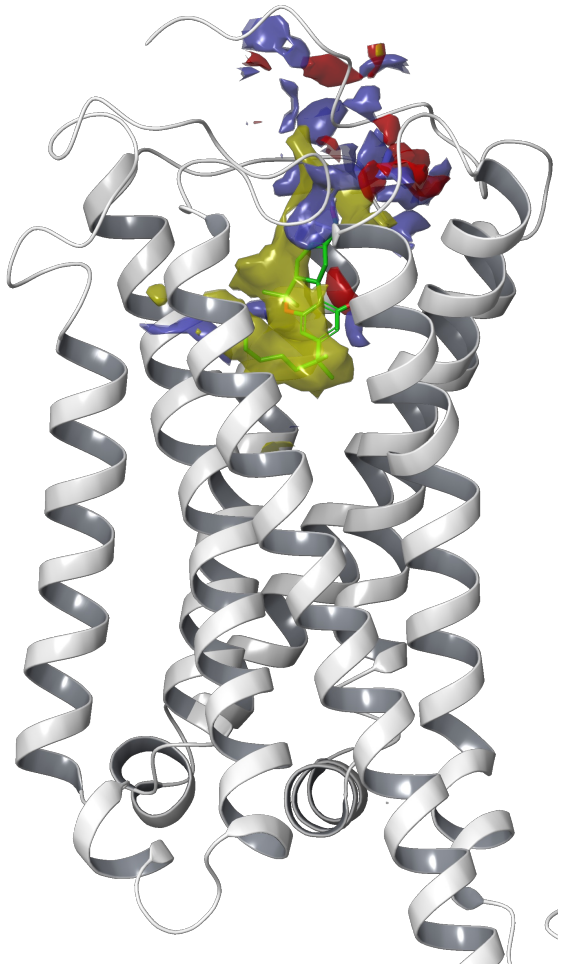
Hit/Ligand optimization
1) 2D similarity searches to identify analogs

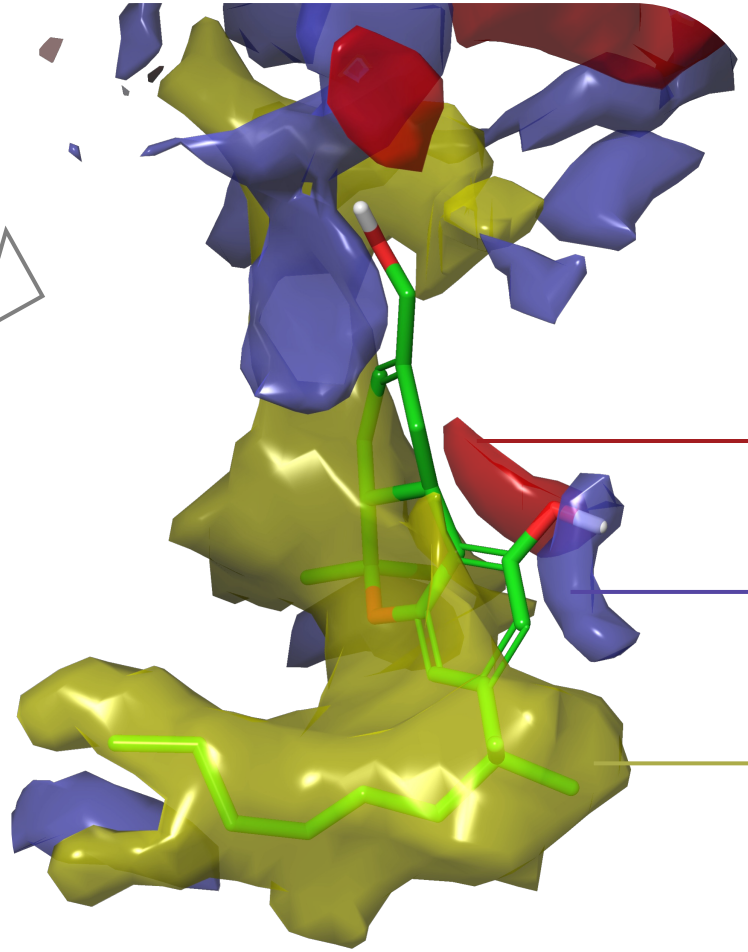
- Analogues can increase selectivity and potency



- Analogs are searched for by:
  - Similar 2D structure
  - Shared substructures
  - This is a lot trickier than it sounds!

Hit/Ligand optimization
2) Binding site analysis using Site Map





Binding site areas

Hydrogen bond acceptor

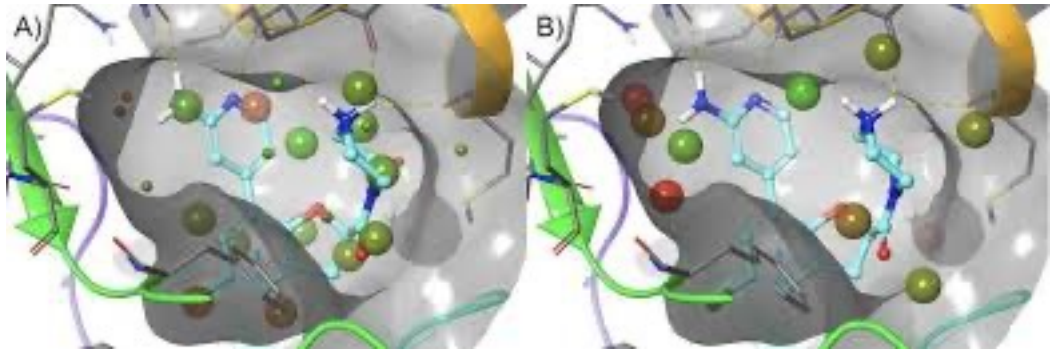
Hydrogen bond donor

Hydrophobic

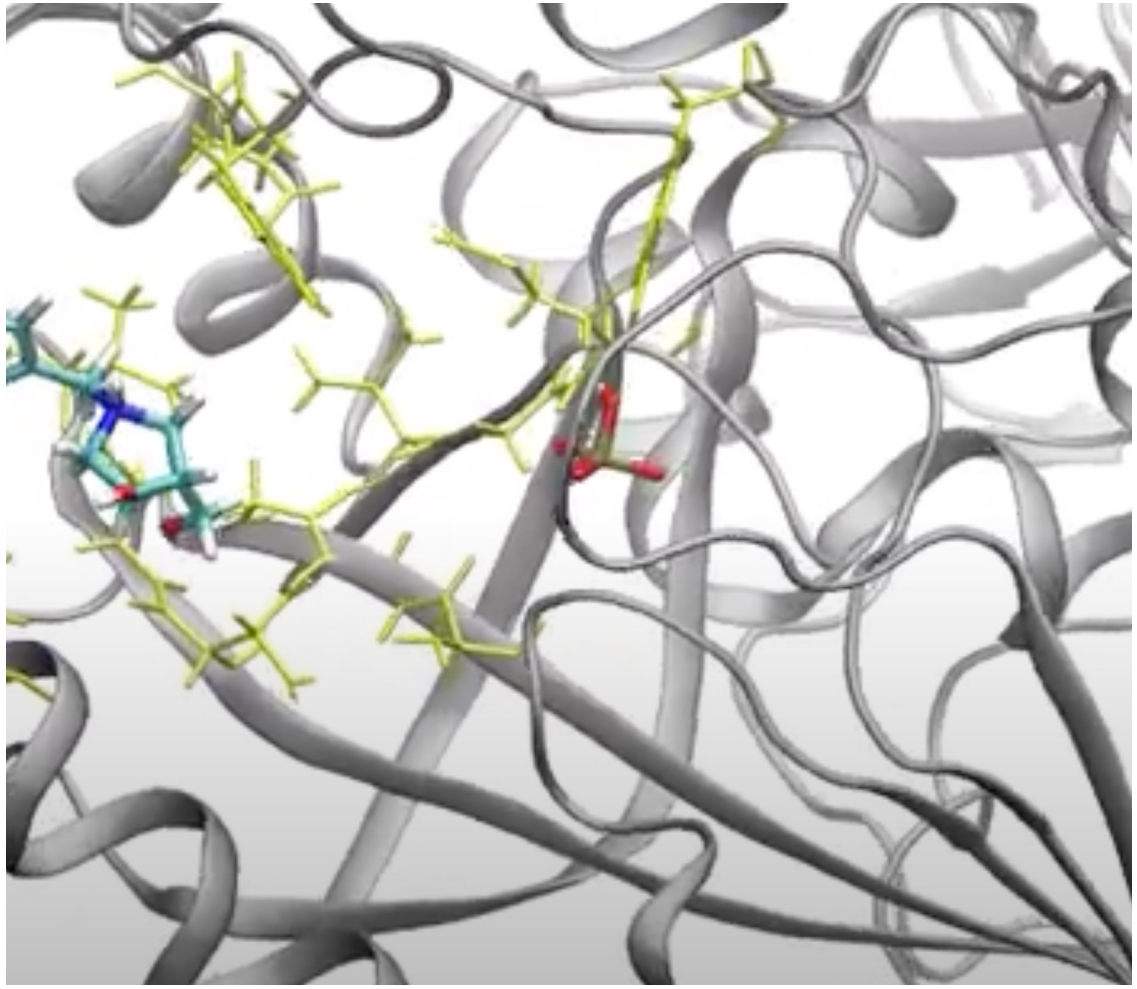
Can be used to guide the design of ligand analogs with more interactions and better fit

Hit/Ligand optimization
3) Water molecule analysis using Water Map

- Stable water
- Loose water



Hit/Ligand optimization
4) Molecular dynamics simulation



Video of ligand entering the binding site of the protein target

https://www.youtube.com/watch?v=ckTqh50r_2w

Molecular dynamics can be used to study ligand-receptor interactions over time.

Computational optimization must be followed by experiments

Computationally modeled

- Binding affinity
- Potency (activity)
- Selectivity / side effects
- Synthetic feasibility
- Lipinski (Pfizer) “Rule of Five“
  - MW < 500
  - log P < 5
  - H donors < 5
  - H acceptors < 10

Experimentally tested

- Toxicity
- Oral bioavailability
- Favourable pharmacokinetics
- Metabolism (e.g. no first pass)
- Elimination pathway/s

Quiz) Which TWO of the statements below are TRUE about virtual screening?

a) Screening large instead of small compound databases typically requires more subsequent optimization of the obtained hit molecules to become lead drugs.

b) Clustering hit molecules (predicted ligands) and then selecting representative molecules of each cluster increases the structural diversity of the molecules selected for purchase. This increases the chance to identify ligands that are active in a biological assay.

c) Virtual screening can only be used for protein targets that already have known ligands.

d) Compound databases only include molecules that have already been synthesized.

e) Ligands with strained poses are automatically scored low in virtual screening.

f) Buying and testing analogs of the first hit molecules can help increase potency and selectivity.

Quiz) Which TWO of the statements below are TRUE about virtual screening?

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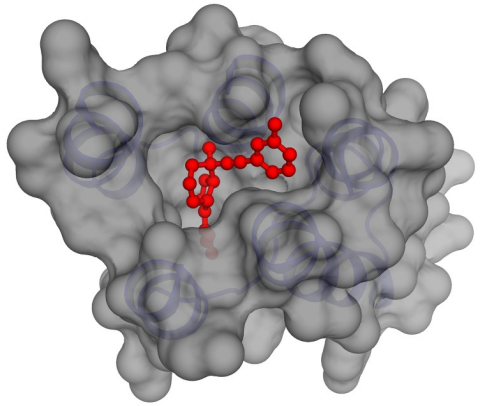
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Lecture learning goals

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- Learn which method to use for which application.

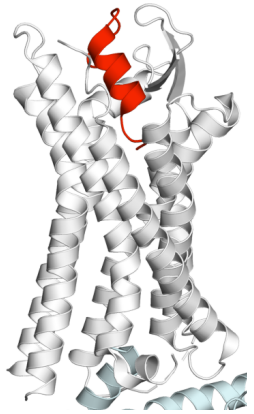
Small molecule ligand

identification and optimization



Protein modeling & peptide ligand

identification and optimization



Breakthroughs in protein modeling and binder design
AlphaFold and RoseTTAFold awards / prizes

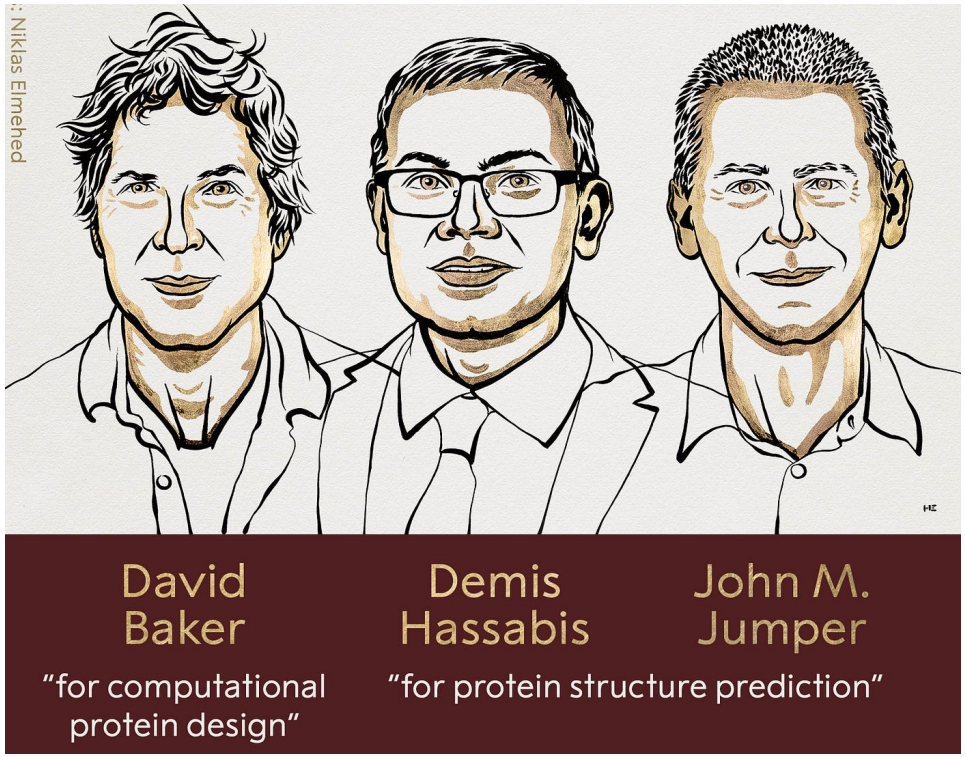
2021

Method of the year by *Nature Methods*

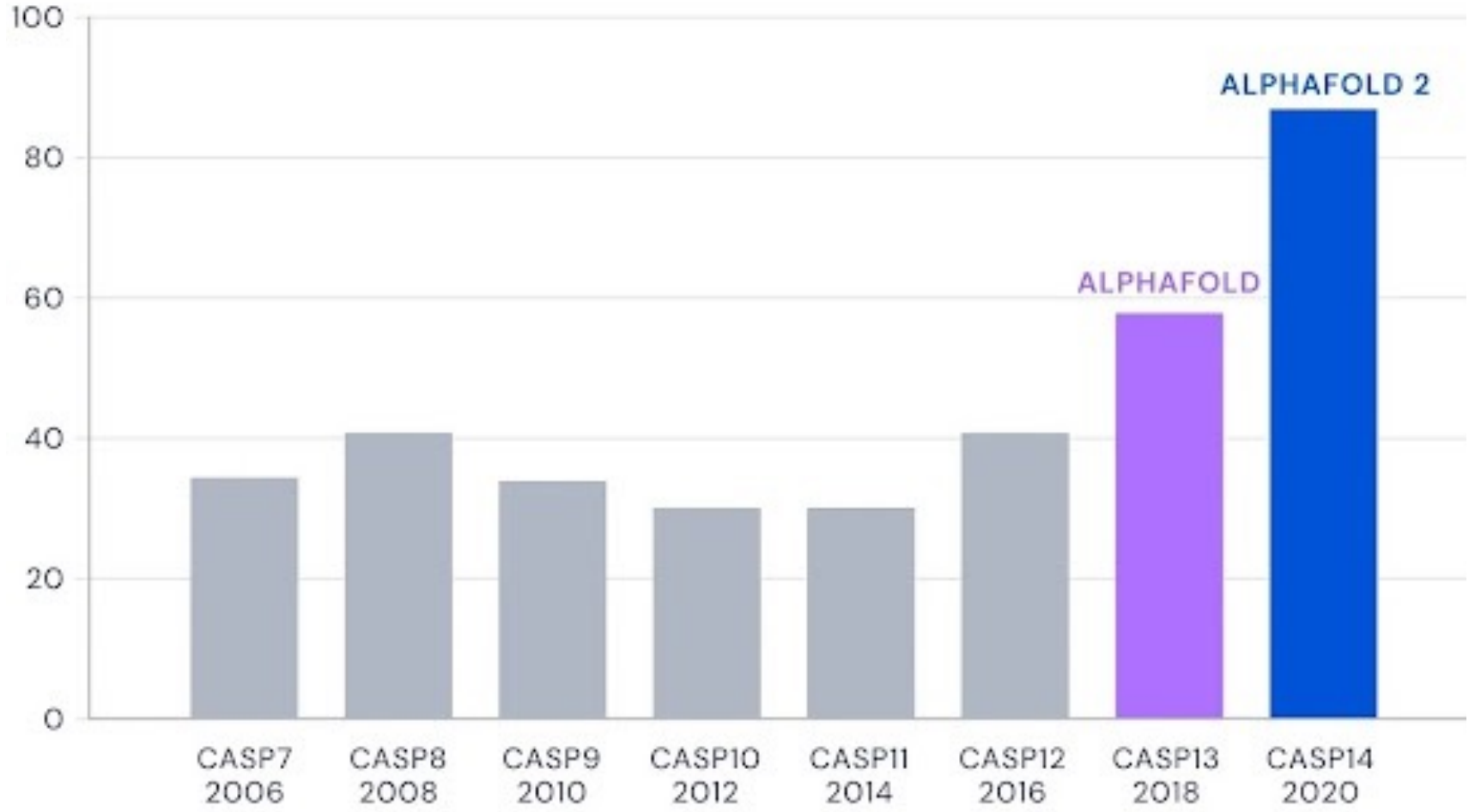
Breakthrough of the year by *Science*

2024

Nobel Prize in Chemistry for protein structure prediction and design



AlphaFold outperformed competition in the CASP modeling competition



Applications: From protein modeling to binder design

Method base	Method	Application
AlphaFold	AlphaFold 1-2	Protein structure modeling
RoseTTA	RoseTTAFold	Protein structure modeling
AlphaFold	AlphaFold (2) -Multimer	Protein/peptide complex modeling
AlphaFold	AlphaFold 3	Protein/peptide or small molecule complex modeling
RoseTTA	RFdiffusion	Protein/peptide design – backbone
RoseTTA	ProteinMPNN	Protein/peptide design – sidechains

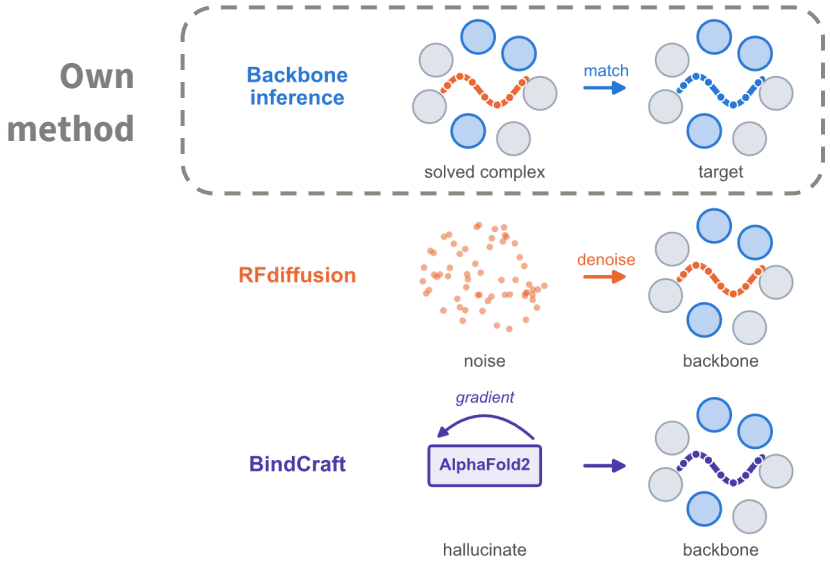
AlphaFold and RosetteFold were the first,

but many additional methods exist e.g., OpenFold, Boltz, Chai …

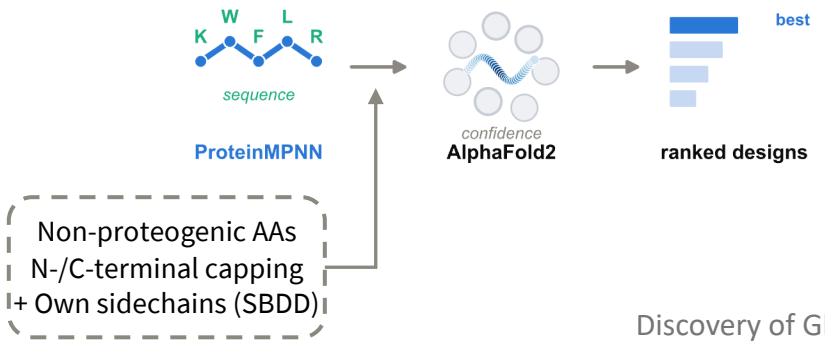
Unpublished project*

Methods

3 backbone design methods

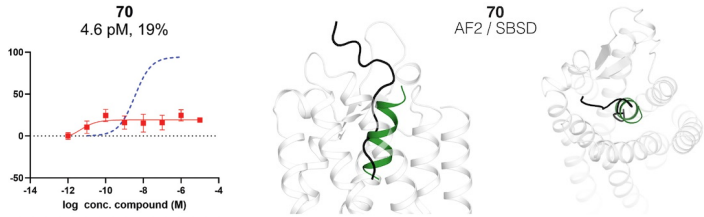


Sidechain selection

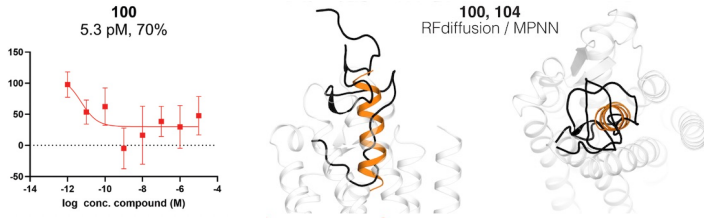


Ligand identification

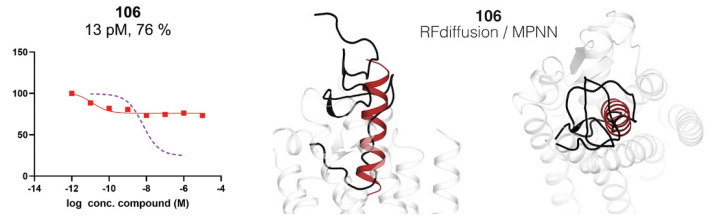
Partial agonists



Antagonists



Inverse agonists



Ligand designers & collaborators

Dept. of Drug Design and Pharmacology

Arman Simonyan



Yashraj Kulkarni



Computer Science (DIKU)

Wouter Boomsma



Biomedicine Institute (BMI)

Alejandro R. Buzón

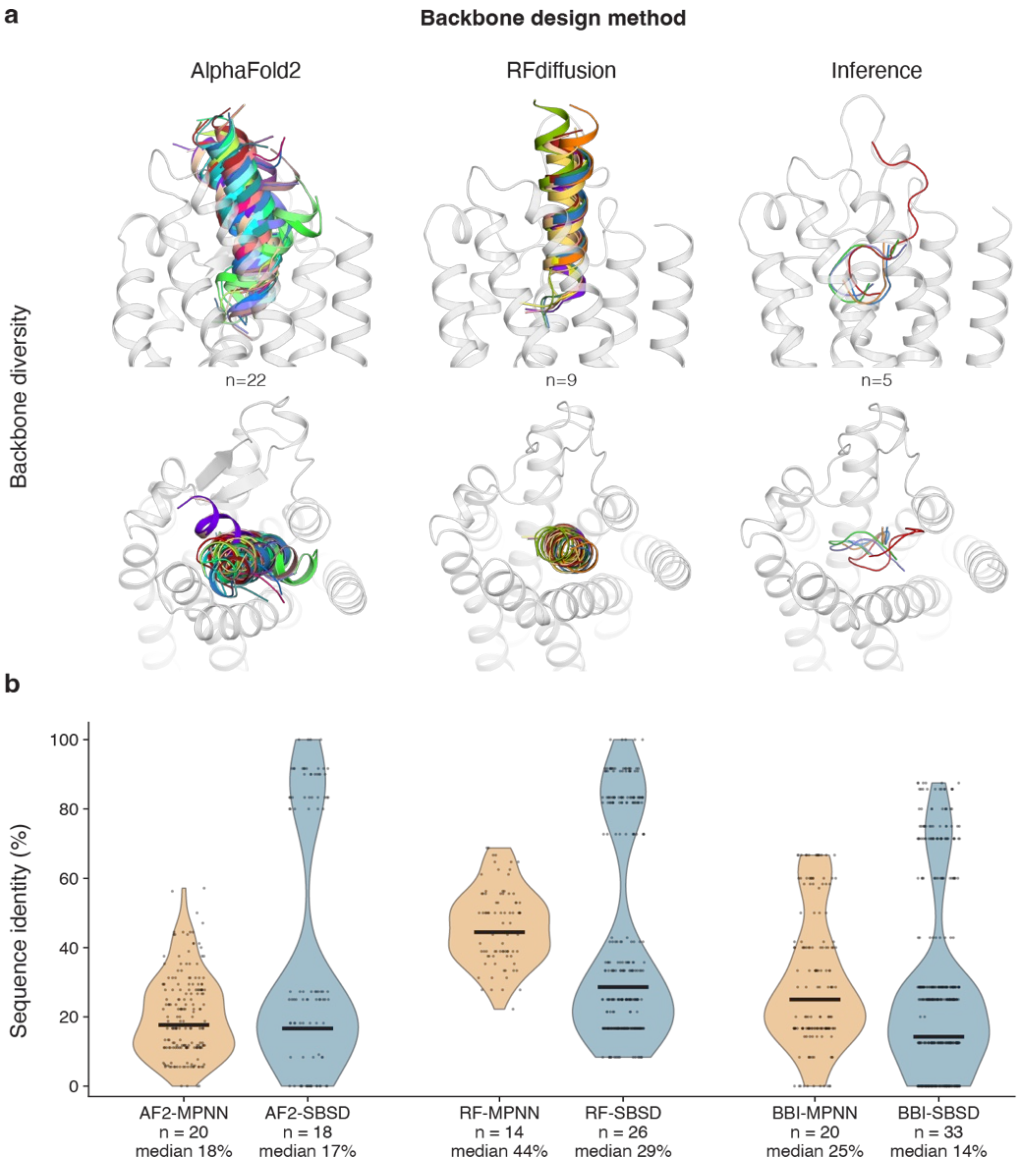


Birgitte Holst



*Simonyan A. et al.,
Discovery of Ghrelin receptor ligands leveraging AlphaFold, Rosetta and human expert designs
In review in *Nature Communications* & draft patent application

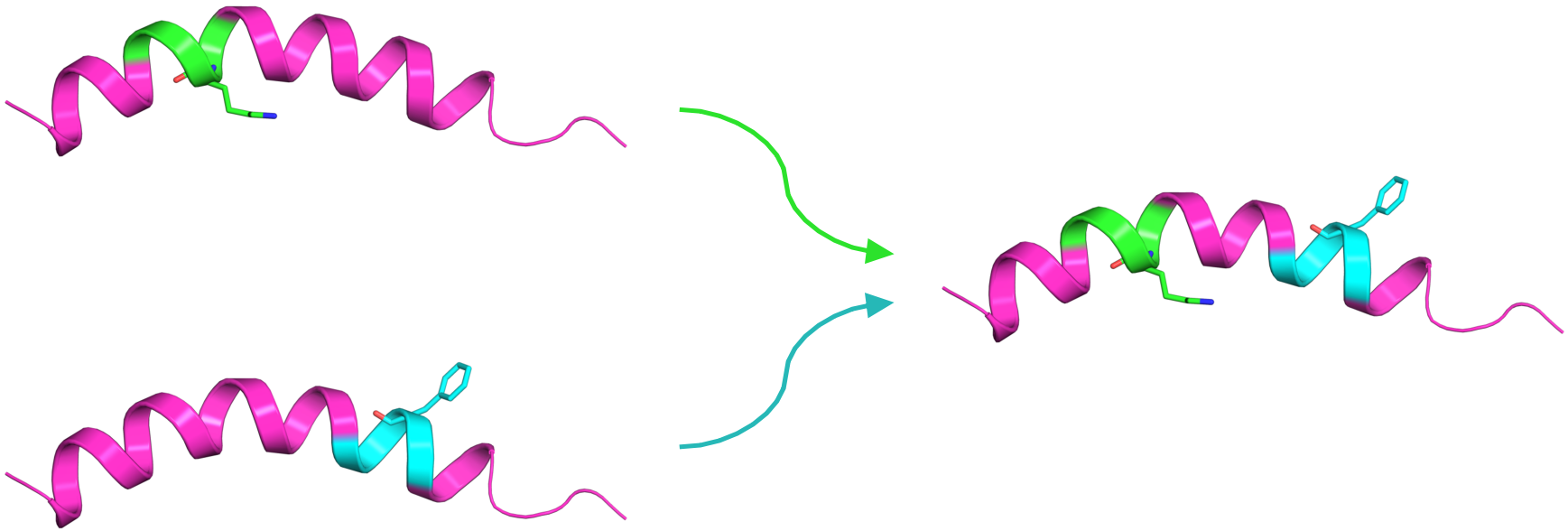
Diversity of peptide backbones and sequences



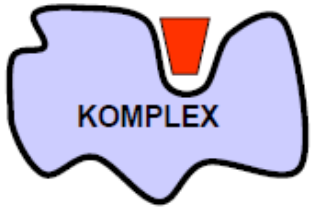
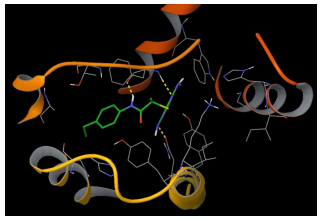
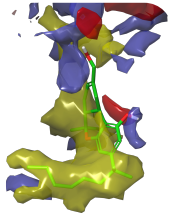
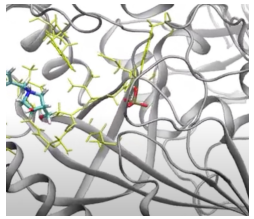
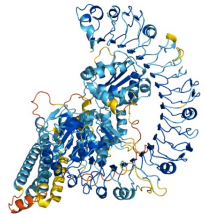
Methods are complementary leading to different mode of action, length and helical/disordered structure

Mode of action	Ligand	Potency	E_{max}	Backbone method	Sidechain method	Len
Partial agonist	34	10 pM	19	RFdiffusion	Structure -based sidechain design	12
	46	1.5 nM	28			12
	69	23 pM	28	AlphaFold2		10
	70	4.7 pM	19			12
Anta-gonist	79	107 pM	84	Inference (disordered)	Protein -MPNN	5
	84	17 pM	76			5
	86	59 pM	110			12
	88	15 pM	92			12
	89	2.2 nM	55			12
	99	40 pM	113	RFdiffusion		16
	100	5.4 pM	70			16
	104	35 pM	33			16
Inverse agonist	102	129 pM	12			17
	106	13 pM	24			18

Peptide ligand optimization
- Combination of two modifications



Summary

	Method	Application
	Docking	Ligand binding mode prediction
	Virtual screening	Ligand/hit identification
	Site map	Ligand/hit optimization
	Molecular dynamics	Ligand/hit optimization (over time)
	AlphaFold / RoseTTA	Protein modeling & binder design

Quiz) Structure-based drug design involves a computational toolbox wherein the tools have complementary applications. Which of the TWO statements below are TRUE?

a) When docking simulates the binding a ligand within its protein target it changes the conformation of both these molecules.

b) Virtual screening is typically used to screen different protein targets for a given ligand.

c) A SiteMap shows which areas in a protein cavity may fit ligand moieties with specific properties, such as hydrophobic, hydrogen bond acceptor and hydrogen bond donor groups. It is therefore often used for structure-based ligand design.

d) AlphaFold can be used to predict protein 3D structures but not to design new ligands.

e) Molecular dynamics simulates the conformational changes and interactions of molecules over time.

f) Today, computational models are as good at predicting ligand toxicity and pharmacokinetics as binding affinity and solubility.

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f) Today, computational models are as good at predicting ligand toxicity and pharmacokinetics as binding affinity and solubility.

Old exam questions

Book sections

Book

The Handbook of Medicinal Chemistry: Principles and Practice.

Ward, Simon E; Davis, Andrew (Eds.)

2th Edition, 2023, The Royal Society of Chemistry England

Sections

4.1 Introduction

4.3 Protein Structures for Structure-based Design

4.5 Ligand-induced Conformational Change